Validation report: PoseidonOS against the SDV challenge brief

Date: 18 October 2026. The evidence below comes from three independent sources:

1. `make check`: lint, no-literal guardrail and the full pytest suite. **481 passed, 18 skipped.** The skips are browser tests that need the optional `playwright` package.
2. `validate.py` (this folder) drives the real engine with expected values computed from the Annex A/B arithmetic, never pasted from engine output. **48 / 48 checks pass.** The full log is `validation_output.txt`. Rerun with `PYTHONPATH=. uv run python deliverables/05_validation/validate.py`.
3. A live run of the dashboard (`make demo`) in a browser, with screenshots in `screenshots/`.

Must-have criteria

Brief requirement	Status	Evidence
Manifest schema encoding all five Annex B cards with the B.0 fields	✅	`schema/guest_manifest.schema.json`, `data/guests/G1–G5.yaml`; MAN-1, MAN-3
Documented way to add a new card	✅	`docs/ADDING_A_CARD.md`, `uv run shipos validate <card>`; MAN-5
Platform budget, all eight classes	✅	`ResourceClass` has 8 members; all eight gauges on screen; PHY-5
Power–heat coupling 1:1, cooling f(T_sea), degraded state	✅	160 → 140 → 120 kW at 12 / 21 / 30 °C, 72 kW degraded at 30 °C; PHY-2, PHY-3
Power reserve f(speed, gensets)	✅	200 / 100 kW; PHY-1
Scenario A end-to-end: verdict ≤ 30 s against envelope, binding named in one sentence	✅	G4 *yes-if* in about 50 ms, binding power.total 101.5 / 100 kW at "slow, 30 °C, one genset down"; G3 *yes-if* on deck.bay1; A-G3, A-G4
…placement, ranked displacement list, one-genset-down statement	✅	5 ranked fixes for G4 and 2 for G3; `holds_one_genset_down` set; A-G4r, A-G3h
Scenario B min 8: decision ≤ 2 s, rationale, cost to loser	✅	< 1 ms; survey: *G3 holds 15 min, 23 survey-min*; SAR: *G3 suspends, 35 kW reserved, 83 survey-min*; B8-*
B: hard constraint first (land ≤ 12 min)	✅	binding time.landing_deadline 41 vs 12 min; B8-hard
B min 12: explicit shedding sequence that keeps the winner alive and states every guest's loss	✅ (see finding F3)	overlap case: survey throttles G5 → 8 kW at t+1 s; SAR defers G3's charge 5 min; G4 never touched; B12-*
Two priority sets → different, coherent outcomes	✅	pause vs suspend at min 8; throttle G5 vs defer G3 at min 12; B8-diff
Identifiable engine D1, D3 + D2/D4 with declared objectives in README	✅ (all four)	`shipos/engine/*.py` docstrings, README table
Scenario C memo ≤ 2 pages; Physical Account	✅	`03_scenario_C_memo/` (2 pp), `02_physical_account/` (1 p + registry)

Should-have criteria

Requirement	Status	Evidence
D2 maximising margin, rejected alternatives shown	✅	lexicographic objective; alternatives with reasons
Jury changes one number live	✅	G4 charging 20 → *yes*; 41 → *yes-if* (107.5 kW); 70 → *yes-if, G5 low-power*; mass 20 t → *no* on hangar mass; UI JSON card editor
Scenario A′ with installation windows	⚠️ partial	port-only → *not now, next port call ≤ 14 days*; 0 crew → logical re-task. The window-forecast path exists (`admission.next_window`), but no Annex card installs at sea, so no real card exercises it
Cooling and deck driven by real public data	✅	Open-Meteo Strait of Sicily cache (`data/env_cache/`), opt-in in the UI, no runtime fetch
Guest exceeding its declaration handled	✅	+10 % warn, +15 % throttle, 60 s shed; `test_overdraw_*`
Scenario C rule at two crew levels, live	✅	14 (human confirm) and 0 (evidence or denial); API acceptance tests
Deterministic, replayable; rationale from the binding constraint	✅	identical SHA-256 over two runs; replay byte-identical; DET-1/2; `rationale.render` only

Constraints

Constraint	Status	Evidence
Offline, laptop, single screen	✅	no remote assets in the UI (OFFL); `--offline` launch

Constraint	Status	Evidence
No clock, randomness or I/O in the engine	✅	PURE check and CLAUDE.md rule 2
Every decision one plain sentence naming the binding constraint	✅	every decision in the log carries `binding` and a rendered rationale
Undocumented assumption is a defect	✅	literal guardrail in CI; 96 registered assumptions with use sites
Civil only; no ML in decisions; no China-based models	✅	rule-based engines; civil cards only

Findings, and what was done about them

#	Finding	Severity	Action
F1	**No card of our own invention.** The brief requires "five cards plus one card of your own invention"	blocker	**Fixed:** `data/guests/G6.yaml` (survivor-care module: safety-critical, requires G5, peak), 11 new registered assumptions, registry regenerated, pinned baseline hashes updated (the replay header serializes every card)
F2	**D2 could put two guests on one Bay 1 half.** The position check counted units per bay group (2 ≤ 2) but not per slot, and the mirror-image dedupe then hid the valid free half. A light half-bay card was admitted onto G5's half	correctness	**Fixed:** `physics/positions.py` adds a per-slot occupancy check (`deck.<slot>.shared`); `engine/placement.py` dedupes only candidates without doubled slots; `engine/rationale.py` renders the new clause. Two regression tests in `tests/physics/test_mass_positions.py` fail before the fix and pass after. The Annex scenarios are unaffected (baseline hash unchanged by the fix)
F3	**Baseline B does not shed at minute 12.** The drone is still airborne (49.5 kW drawn), so the honest answer is "no shedding needed" plus the armed ladder (G5 throttle → G1 off → G3 standby). Even the brief's own numbers (35 + 45 + 2.5 = 82.5 kW < 100) do not overload	demo risk	Not changed. The engine is right. **In the demo, narrate the armed ladder and then show the charging-overlap case** (106.5 kW) from `validation_output.txt` (B12-sar / B12-survey-on-overlap) or `uv run pytest tests/engine/test_shedding.py -v`
F4	**UI trap.** *Run the mission* keeps *Situation = In port* (Scenario A), so *Report battery* fails silently (HTTP 422: G4 is not aboard)	demo risk	Not changed (UI is outside this task). **Runbook step: select "At sea, running a mission" first**
F5	**Stale submission docs.** Docs were pinned to `fcd8c58`, said "G6 is absent", used the old name ShipOS, gave Windows-only launch steps, and lacked "another week" and the Annex C closing question	submission	**Fixed:** new README (≤ 3 pp), Physical Account in Annex C form with the closing question, memo restructured to the brief's questions, `make demo`
F6	**Brief arithmetic.** "Any four exceed the 10 t margin" is false for {G1,G2,G4,G5} (7.61 t) and {G1,G3,G4,G5} (9.61 t)	note	Recorded in the README and Physical Account; the engine follows the card masses
F8	**Response-time gauge always empty.** Nothing timed the engine; the gauge defaulted to 0 s and the server hid it	demo gap	**Fixed:** the API adapter times every real engine call (`time.perf_counter`, outside the pure engine) and the gauge shows the latest one against its A.8 ceiling (admission 30 s, arbitration 2 s, first shed 5 s, including the scheduled 1 s shed offset). Measured live: 33–49 ms admission, 0.5 ms D3, 0.2 ms D4. Latency stays out of the hashed log, and the pinned replay hashes are unchanged
F7	**A.11 portable ≤ 250 kg** is declared in ship data but not enforced, and no card installs at sea	gap	Listed under limitations and "another week"

Files changed in the repository by this validation (uncommitted)

`data/guests/G6.yaml` (new), `data/assumptions/other.yaml`, `data/assumptions/compute_time.yaml`, `docs/PHYSICAL_ACCOUNT_REGISTRY.md` (regenerated), `shipos/physics/positions.py`, `shipos/engine/placement.py`, `shipos/engine/rationale.py`, `tests/physics/test_mass_positions.py`, three pinned hashes in `tests/sim/`, the same hash in `docs/SIM_EXECUTION.md`, `docs/ENVIRONMENT.md` and `shipos/sim/README.md`, plus `README.md` and `Makefile` (`demo` target). For F8: `shipos/api/adapter.py`, `shipos/api/server.py`, `shipos/api/static/app.js`, `shipos/physics/budget.py`, `tests/api/test_server.py`, `tests/physics/test_budget.py`.